

SIMULATION OF SAWTEETH INSTABILITIES IN COMPASS

by

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*We are nothing but space dust trying to find its way back
to the stars* - David Jones

Abstract

Being able to control the plasma state and its evolution in real time is crucially important for magnetic confinement nuclear fusion stability. However, the complex physics describing the plasma state requires an extensive amount of computational power and a long computational time. An alternative for this conundrum is to develop a model that simplifies the physics, lowering the running time.

Experimental data combined with a model that predicts the transport behaviour of a tokamak (COMPASS), allows to accurately predict the state of the system. An accurate prediction from the plasma state allows the development and integration of a closed loop control system.

A simulation of a sawtooth instability (plasma instability) using real gathered data from the COMPASS's sensors; is developed in this thesis. This simulation uses a 1D Rapid Plasma Transport (RAPTOR) code.

This code solves two coupled 1D partial differential diffusion equations for the electron temperature and the poloidal magnetic flux. The RAPTOR code combined with collected data allowed to calculate the critical magnetic shear for COMPASS tokamak.

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Symbols

Re	Reynolds Number
ρ	density $\left[\frac{M}{L^3}\right]$
μ	dynamic viscosity $\left[\frac{M}{LT}\right]$
ν	kinematic viscosity $\left[\frac{L^2}{T}\right]$
L_c	characteristic length
p	pressure
κ	thermal conductivity
E	total energy
τ_{ij}	shear stress tensor $\left[\frac{M}{LT^2}\right]$
S_{ij}	strain-rate tensor $\left[\frac{1}{T}\right]$
Ω_{ij}	rotation-rate tensor
f_i	body force
ϵ	dissipation rate
η	Kolmogorov length scale
τ	Kolmogorov time scale
v	Kolmogorov velocity scale
$f(r, t)$	Longitudinal velocity autocorrelation function
$L_{11}(t)$	Longitudinal integral scale
$\lambda_f(t)$	Longitudinal Taylor microscale
$g(r, t)$	Transverse velocity autocorrelation function
$L_{22}(t)$	Transverse integral scale

$\lambda_g(t)$	Transverse Taylor microscale
δ	velocity boundary layer thickness
δ_T	thermal boundary layer thickness
Pr	Prandlt Number
$()^R$	resolved scale quantity
$()^M$	modeled scale quantity (subgrid scale quantity)
$\overline{()}$	mean quantity
$()'$	fluctuation quantity
$()_{DNS}$	quantitiy from DNS
M_t	turbulent mach number
a	speed of sound
k	kinetic energy
u^{sol}	solenoidal field
u^{dil}	dilatational field

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Chapter 1

Introduction

Chapter 2

Modeling

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Appendices